

Sensor Inspection and Cable Repair — Conveyor

Signal Quality Fault — Burst Dropout on Vibration Channel

Asset: AST_CON_001 (Conveyor Assembly) | Channel: CH_31B | LINE_001

Bearing Type: SKF22212 | Est. Duration: 2–3 h | Crew: 1 (OT Specialist) | Priority: HIGH

SOP_004 | v1.0 | Revised: 2026-04-01

1. Purpose and Scope

This procedure covers inspection and repair of vibration sensor channels on the conveyor assembly (AST_CON_001) where the Data Foundation Agent has raised a `signal_quality_alert` indicating burst-pattern dropouts. Burst dropouts — consecutive null readings in clusters of 2 to 4 — reliably indicate a physical connection fault (loose terminal or cable damage), NOT a bearing fault. This SOP does not cover bearing replacement. A visual bearing inspection is included in the scope while access is available. CRITICAL: AST_CON_001 LOTO permit LOTO_003 expired on 2026-04-15. A renewed permit MUST be issued before any work starts. This SOP will be blocked by the Safety Agent if `LOTO_003 is_current = false`.

2. Prerequisites

- CRITICAL: LOTO_003 must be renewed before work starts. Confirm `is_current = true` and `expiry > today`.
- Signal quality alert confirmed: burst-pattern dropout on CH_31B (historian tag: `PI:CON001.BRG9:VIB`)
- Dropout percentage confirmed: $\geq 1\%$ over 7-day window in burst pattern
- Consecutive null records confirmed at TEL_0015 and TEL_0016 (`signal_quality_score < 0.20`)
- Work order raised with priority HIGH (data integrity risk on bottleneck asset)
- OT instrument specialist assigned (sensor cable repair qualification required)
- Access to junction box J-301 confirmed
- Torque screwdriver and replacement cable/connector from instrument store

3. Safety Requirements

3.1 LOTO Isolation Procedure

LOTO Permit	LOTO_003 — MUST be renewed before this SOP can proceed
Isolation Points	MCC-20, GB-301
Energy Types	Electrical, Mechanical
SAFETY STOP	If LOTO_003 <code>is_current = false</code> : DO NOT PROCEED. Obtain renewed permit. The Safety Agent will block this work order until permit is current.
Verification	After isolation: confirm conveyor does not move. Confirm zero voltage at MCC-20 terminals.

3.2 PPE and Zone Requirements

Mandatory PPE	Safety helmet, electrical insulation gloves, safety boots
Zone — AST_CON_001	B — Unrestricted approach with standard PPE. Conveyor must be fully isolated before accessing belt or bearing zones.

4. Tools and Materials Required

No bearing parts required for this SOP. Cable components from instrument store:

- Replacement signal cable (2-conductor shielded, 1 m minimum) — from OT instrument store
- Replacement terminal block connector (M4 screw terminal, rated for signal use)
- Torque screwdriver with 5–6 Nm range (CRITICAL: terminal tightening must be to spec)

- Multimeter for continuity and insulation resistance check
- Cable tie kit
- Label printer or cable marker for re-labelling repaired cable
- Torch or work light for junction box inspection

5. Step-by-Step Procedure

1	STOP CHECK: Confirm LOTO_003 is_current = true in safety system before touching any equipment.
2	Apply LOTO per Section 3.1. Verify conveyor is stationary and MCC-20 is locked.
3	Locate junction box J-301 (non-drive end, conveyor frame).
4	Open J-301 cover. Visually inspect all terminal screws for the CH_31B signal conductor.
5	Attempt to hand-wiggle each conductor at the terminal. Any movement indicates a loose termination.
6	If loose terminal found: re-terminate the cable conductor into the terminal block. Tighten terminal screw to 5–6 Nm using calibrated torque screwdriver.
7	If cable damage found (fraying, insulation breach, cut): cut out damaged section and install replacement cable, re-terminate both ends to 5–6 Nm.
8	Perform continuity check with multimeter: confirm signal path from sensor to historian connection point.
9	Perform insulation resistance check: confirm IR > 1 MΩ between signal conductor and shield/ground.
10	Label repaired cable/terminal with date and technician ID.
11	Close J-301 cover and secure all fasteners.
12	While access is available: perform visual inspection of BRG_009 (DRIVE bearing). Note any visible damage, grease leakage, or discolouration. Record findings.
13	Also inspect BRG_010 (NON_DRIVE bearing) if accessible. Record condition.
14	Remove LOTO. Controlled conveyor restart.
15	Monitor CH_31B historian tag for 15 consecutive clean readings with signal_quality_score ≥ 0.97.
16	If signal quality restored: QA PASS. Record on work order and close.
17	If dropouts continue: escalate — cable damage may extend beyond J-301 or sensor itself may need replacement.

6. Critical Specifications

Terminal Tightening Torque	5–6 Nm — MUST use torque screwdriver. Hand-tightening is the root cause of CASE_003 recurring fault.
Insulation Resistance	≥ 1 MΩ — signal conductor to shield/ground
Signal Quality Pass Criterion	≥ 0.97 for 15 consecutive readings — no burst dropout in monitoring window
Channel ID — do not rename	CH_31B (historian tag: PI:CON001.BRG9:VIB) — ID is fixed in telemetry schema. Renaming will break downstream agent routing.

7. Post-Repair Quality Acceptance Criteria

Parameter	Criterion	Failure Action
Signal Quality Score (CH_31B)	≥ 0.97 for 15 consecutive readings	Continue investigation — escalate to sensor replacement
Dropout Rate (7-day window)	< 0.1% (zero burst events)	Do not close work order — continue monitoring
Bearing Vib at restart (BRG_009)	≤ 2.8 mm/s	Raise separate bearing inspection case
Bearing Temp at restart (BRG_009)	≤ 58°C	Raise separate bearing inspection case

Parameter	Criterion	Failure Action
BRG_010 visual condition	No visible damage or leakage	Raise separate inspection case

8. Common Issues and Troubleshooting

- If dropout continues after terminal re-tightening → cable insulation breach along run from J-301 to sensor. Replace full cable run.
- If sensor itself is damaged (connector cracked, IP rating breached) → sensor replacement required. Order OEM sensor and raise separate work order.
- If bearing inspection reveals unexpected vibration or discolouration → raise separate bearing inspection case under SOP_006.
- If LOTO_003 is still expired at next inspection → LOTO renewal is the safety team's responsibility; escalate to maintenance supervisor.
- Terminal torque below 5 Nm will allow re-loosening from conveyor vibration — always use torque screwdriver, not electric driver.

9. Related Documents

- SOP_006 — Post-Repair QA Procedure (all assets)
- LOTO_003 permit record — AST_CON_001 (MUST be renewed, expiry: 2026-04-15)
- CASE_003 — Sensor dropout, CH_31B, loose terminal at J-301, LOTO permit expired
- CASE_011 — Preventive check, AST_CON_001, BRG_010 (NON_DRIVE), no fault